

Technical Report QA Report

Date: 2026-09-13

Checks completed

- `validate_python_comments.py scripts/build_technical_report.py`: passed.
- Process flow, parameter, and validation figures were rendered as PNG and inspected at high resolution.
- The process-flow redraw removed the overlapping middle title and redundant feedback labels.
- All axes carry units or dimensionless labels; CO_2 , N_2 , CH_4 , and H_2 use readable subscripts.
- Isotherm markers identify the current feed partial-pressure points; no experimental points or uncertainty bars were invented.
- The IDAES reference and reduced-order evidence are explicitly separated in the report.
- SVG outputs retain editable text; PDF outputs are single-page exports with TrueType font settings.
- Raw YAML, CSV, and JSON result bundles remain separate from derived parameter CSV files.
- `python scripts/render_reports.py`: passed; the main technical report rendered to a seven-page PDF and its figure plan, method-search packet, and QA report each rendered to one-page PDFs.
- The report contact sheet and representative page PNGs were inspected after PDF rendering; formulas, Chinese text, tables, embedded figures, and the Aspen-like flowchart were legible with no visible right-edge clipping.
- The PDF wrapper uses XeLaTeX with a Unicode-capable text font, explicit Markdown math extensions, and a maximum-width table wrapper. Temporary TeX files are removed; the committed render manifest contains repository-relative paths only.

Scientific boundary

The report validates implementation behavior, units/bounds, numerical termination, residuals, mass balances, CSS, and a limited fixed-bed mesh change. It does not validate material data, kinetics, membrane permeance, thermal behavior, a full distributed PSA, or industrial process performance against experiments, literature, or native Aspen results.